

$$\begin{aligned}
\begin{Bmatrix} \Delta F_{x_b} + \Delta W_{x_b} \\ \Delta F_{y_b} + \Delta W_{y_b} \\ \Delta F_{z_b} + \Delta W_{z_b} \end{Bmatrix} &= \begin{Bmatrix} F_{x_b,u} & 0 & F_{x_b,w} \\ 0 & F_{y_b,v} & 0 \\ F_{z_b,u} & 0 & F_{z_b,w} \end{Bmatrix} \begin{Bmatrix} \Delta u \\ \Delta v \\ \Delta w \end{Bmatrix} + \begin{Bmatrix} 0 & F_{x_b,q} & 0 \\ F_{y_b,p} & 0 & F_{y_b,r} \\ 0 & F_{z_b,q} & 0 \end{Bmatrix} \begin{Bmatrix} \Delta p \\ \Delta q \\ \Delta r \end{Bmatrix} \\
&+ \begin{Bmatrix} F_{x_b,\dot{u}} & 0 & F_{x_b,\dot{w}} \\ 0 & 0 & 0 \\ F_{z_b,\dot{u}} & 0 & F_{z_b,\dot{w}} \end{Bmatrix} \begin{Bmatrix} \Delta \dot{u} \\ \Delta \dot{v} \\ \Delta \dot{w} \end{Bmatrix} + \begin{Bmatrix} 0 & F_{x_b,\delta_e} & 0 \\ F_{y_b,\delta_a} & 0 & F_{y_b,\delta_r} \\ 0 & F_{z_b,\delta_e} & 0 \end{Bmatrix} \begin{Bmatrix} \Delta \delta_a \\ \Delta \delta_e \\ \Delta \delta_r \end{Bmatrix} \\
&+ \begin{Bmatrix} 0 & -W \cos \theta_o & 0 \\ W \cos \theta_o & 0 & 0 \\ 0 & -W \sin \theta_o & 0 \end{Bmatrix} \begin{Bmatrix} \Delta \phi \\ \Delta \theta \\ \Delta \psi \end{Bmatrix} \quad (7.5.11)
\end{aligned}$$

and the small-disturbance moment approximations become

$$\begin{aligned}
\begin{Bmatrix} \Delta M_{x_b} \\ \Delta M_{y_b} \\ \Delta M_{z_b} \end{Bmatrix} &= \begin{Bmatrix} 0 & M_{x_b,v} & 0 \\ M_{y_b,u} & 0 & M_{y_b,w} \\ 0 & M_{z_b,v} & 0 \end{Bmatrix} \begin{Bmatrix} \Delta u \\ \Delta v \\ \Delta w \end{Bmatrix} + \begin{Bmatrix} M_{x_b,p} & 0 & M_{x_b,r} \\ 0 & M_{y_b,q} & 0 \\ M_{z_b,p} & 0 & M_{z_b,r} \end{Bmatrix} \begin{Bmatrix} \Delta p \\ \Delta q \\ \Delta r \end{Bmatrix} \\
&+ \begin{Bmatrix} 0 & 0 & 0 \\ M_{y_b,\dot{u}} & 0 & M_{y_b,\dot{w}} \\ 0 & 0 & 0 \end{Bmatrix} \begin{Bmatrix} \Delta \dot{u} \\ \Delta \dot{v} \\ \Delta \dot{w} \end{Bmatrix} + \begin{Bmatrix} M_{x_b,\delta_a} & 0 & M_{x_b,\delta_r} \\ 0 & M_{y_b,\delta_e} & 0 \\ M_{z_b,\delta_a} & 0 & M_{z_b,\delta_r} \end{Bmatrix} \begin{Bmatrix} \Delta \delta_a \\ \Delta \delta_e \\ \Delta \delta_r \end{Bmatrix} \quad (7.5.12)
\end{aligned}$$

Notice that within the accuracy of small-disturbance theory, ΔF_{x_b} , ΔF_{z_b} , ΔW_{x_b} , ΔW_{z_b} , and ΔM_{y_b} depend only on $\Delta \dot{u}$, $\Delta \dot{w}$, Δu , Δw , Δq , $\Delta \delta_e$, and $\Delta \theta$, while ΔF_{y_b} , ΔW_{y_b} , ΔM_{x_b} , and ΔM_{z_b} depend only on Δv , Δp , Δr , $\Delta \delta_a$, $\Delta \delta_r$, and $\Delta \phi$.

Rearranging Eqs. (7.5.11) and (7.5.12), we have

$$\begin{aligned}
\begin{Bmatrix} \Delta F_{x_b} + \Delta W_{x_b} \\ \Delta F_{z_b} + \Delta W_{z_b} \\ \Delta M_{y_b} \end{Bmatrix} &= \begin{Bmatrix} F_{x_b,\dot{u}} & F_{x_b,\dot{w}} & 0 \\ F_{z_b,\dot{u}} & F_{z_b,\dot{w}} & 0 \\ M_{y_b,\dot{u}} & M_{y_b,\dot{w}} & 0 \end{Bmatrix} \begin{Bmatrix} \Delta \dot{u} \\ \Delta \dot{w} \\ \Delta \dot{q} \end{Bmatrix} + \begin{Bmatrix} F_{x_b,u} & F_{x_b,w} & F_{x_b,q} \\ F_{z_b,u} & F_{z_b,w} & F_{z_b,q} \\ M_{y_b,u} & M_{y_b,w} & M_{y_b,q} \end{Bmatrix} \begin{Bmatrix} \Delta u \\ \Delta w \\ \Delta q \end{Bmatrix} \\
&+ \begin{Bmatrix} F_{x_b,\delta_e} \\ F_{z_b,\delta_e} \\ M_{y_b,\delta_e} \end{Bmatrix} \Delta \delta_e + \begin{Bmatrix} -W \cos(\theta_o) \\ -W \sin(\theta_o) \\ 0 \end{Bmatrix} \Delta \theta \quad (7.5.13)
\end{aligned}$$

$$\begin{aligned}
\begin{Bmatrix} \Delta F_{y_b} + \Delta W_{y_b} \\ \Delta M_{x_b} \\ \Delta M_{z_b} \end{Bmatrix} &= \begin{Bmatrix} F_{y_b,v} & F_{y_b,p} & F_{y_b,r} \\ M_{x_b,v} & M_{x_b,p} & M_{x_b,r} \\ M_{z_b,v} & M_{z_b,p} & M_{z_b,r} \end{Bmatrix} \begin{Bmatrix} \Delta v \\ \Delta p \\ \Delta r \end{Bmatrix} + \begin{Bmatrix} F_{y_b,\delta_a} & F_{y_b,\delta_r} \\ M_{x_b,\delta_a} & M_{x_b,\delta_r} \\ M_{z_b,\delta_a} & M_{z_b,\delta_r} \end{Bmatrix} \begin{Bmatrix} \Delta \delta_a \\ \Delta \delta_r \end{Bmatrix} \\
&+ \begin{Bmatrix} W \cos(\theta_o) \\ 0 \\ 0 \end{Bmatrix} \Delta \phi \quad (7.5.14)
\end{aligned}$$